4. System Analysis & Design

4.1 Problem Statement & Objectives

4.1.1 Problem Statement

Generative AI and automation are changing which jobs exist, what they pay and which skills they demand, yet the evidence is scattered across reports, surveys and datasets that few stakeholders can read together. Job seekers cannot tell whether their occupation is exposed. Employers cannot benchmark workforce risk against their industry. Policy makers lack a single, consistent view of displacement and reskilling needs. Analysts spend more time cleaning and reconciling data than interpreting it.

This project addresses that gap with an interactive Business Intelligence system, "AI Impact on Jobs Market". It consolidates 30,000 structured records (25 columns, 2024–2026) into a Star Schema model and exposes AI exposure, displacement risk, automation trends, skill requirements and labor market metrics through a four-page Power BI report.

4.1.2 Project Objectives

1. **Consolidate** AI-and-labor-market data into one validated, analysis-ready Star Schema model.
2. **Quantify** AI exposure, automation risk and estimated jobs affected by industry, job category, geography and year.
3. **Compare** salary levels and salary change against automation risk to show economic impact.
4. **Identify** in-demand skills and reskilling priorities for jobs with high exposure.
5. **Enable self-service analysis** through slicers, cross-filtering and drill-through, without writing queries.
6. **Document** a repeatable pipeline (generation, cleaning, modeling, reporting) that another analyst can rebuild and extend.

4.1.3 Scope and Assumptions

- The dataset is a **synthetic, pre-processed CSV** generated with Python and Claude; it contains no personal data and is not a live feed.
- The analytical period is 2024–2026; all figures are illustrative and are not official labor statistics.

- The deliverable is a Power BI Desktop report (.pbix); publishing to Power BI Service is an optional extension (see 4.6.3).

4.1.4 Use Case Diagram

Four actors interact with the system. Three consume the dashboard; the Project Analyst also maintains the data pipeline and the semantic model.

```
flowchart LR
  JS(["Job Seeker"])
  HR(["HR / Employer"])
  PM(["Policy Maker"])
  PA(["Project Analyst"])
  subgraph SYS["AI Impact on Jobs Market – Power BI System"]
    UC1(("UC1 View Executive Summary"))
    UC2(("UC2 Explore Industry Analysis"))
    UC3(("UC3 Assess Risk and Automation"))
    UC4(("UC4 Review Skills and Training"))
    UC5(("UC5 Apply Interactive Filters"))
    UC6(("UC6 Drill Down / Drill Through"))
    UC7(("UC7 Export and Share Insights"))
    UC8(("UC8 Load and Refresh Data"))
    UC9(("UC9 Clean and Transform Data"))
    UC10(("UC10 Maintain Model and DAX"))
    UC11(("UC11 Validate Data Quality"))
  end
  JS --- UC3
  JS --- UC4
  JS --- UC5
  HR --- UC2
  HR --- UC3
  HR --- UC4
  HR --- UC5
  HR --- UC6
  PM --- UC1
  PM --- UC2
  PM --- UC3
  PM --- UC7
  PA --- UC5
  PA --- UC6
  PA --- UC8
  PA --- UC9
  PA --- UC10
```

```
  PA --- UC11
  UC1 -. include .-> UC5
  UC2 -. include .-> UC5
  UC3 -. include .-> UC5
  UC4 -. include .-> UC5
  UC8 -. include .-> UC9
  UC10 -. include .-> UC11
```

4.1.5 Actor Descriptions

Actor	Role	Main goals	Dashboard pages used
Job Seeker	Individual evaluating career options	Check automation risk of a job category; find skills worth learning	Risk & Automation, Skills & Training
HR / Employer	Workforce planner at a company	Benchmark exposure by industry and company size; plan reskilling	Industry Analysis, Risk & Automation, Skills & Training
Policy Maker	Government or institutional decision-maker	Understand total jobs affected by region and industry; design interventions	Executive Summary, Industry Analysis, Risk & Automation
Project Analyst	Builder and maintainer of the system	Load, clean, model, validate and extend the data and report	All pages, plus Power Query and model view

4.1.6 Use Case Descriptions

ID	Use case	Primary actor	Precondition	Main flow	Postcondition
UC1	View Executive Summary	Policy Maker	Report is open and data is loaded	Opens the Executive Summary page; reads KPI cards (jobs affected, average automation risk, average salary) and trend visuals	Actor has a high-level view of AI impact for 2024–2026

ID	Use case	Primary actor	Precondition	Main flow	Postcondition
UC2	Explore Industry Analysis	HR / Employer, Policy Maker	Model relationships are active	Selects an industry or company size; compares jobs affected and exposure across industries	Industry-level comparison is displayed
UC3	Assess Risk and Automation	Job Seeker, HR / Employer, Policy Maker	Risk measures exist in the model	Filters by job category or region; inspects automation risk score and risk-level distribution	Actor identifies high-risk occupations
UC4	Review Skills and Training	Job Seeker, HR / Employer	Dim_Job holds skill attributes	Selects a job category; reviews required skills and skill levels; compares to risk	Actor identifies reskilling priorities
UC5	Apply Interactive Filters	All actors	A report page is open	Chooses values in slicers (year, industry, country, job category); all visuals recalculate through the model	Visuals reflect the selected filter context
UC6	Drill Down / Drill Through	HR / Employer, Project Analyst	A hierarchy or drill-through page is defined	Drills from industry to company or from region to country; opens detail page for a selection	Detailed records for the selection are shown
UC7	Export and Share Insights	Policy Maker	Report is open	Exports a visual or page to PDF/PowerPoint or data to CSV; shares the report file	Insight is available outside Power BI
UC8	Load and Refresh Data	Project Analyst	CSV file exists at the configured path	Points Power BI to the CSV; runs Refresh	Fact and dimension tables are populated
UC9	Clean and Transform Data	Project Analyst	Raw CSV is loaded in Power Query	Applies typing, trimming, null handling, deduplication; splits flat data into fact and dimensions; creates keys	Clean, keyed tables are ready for modeling

ID	Use case	Primary actor	Precondition	Main flow	Postcondition
UC10	Maintain Model and DAX	Project Analyst	Tables are loaded	Defines relationships, hierarchies and DAX measures; formats fields	Semantic model supports all report pages
UC11	Validate Data Quality	Project Analyst	Model is built	Runs row-count, key-integrity and measure checks (see 4.6.2)	Validation results are recorded; defects fixed

4.1.7 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall import the 30,000-record, 25-column CSV dataset into Power BI through Power Query.	Must
FR-02	The system shall clean the data in Power Query: set data types, trim text, standardize categories, handle nulls and remove duplicates.	Must
FR-03	The system shall split the cleaned data into Fact_AI_Job_Market and four dimension tables (Dim_Company, Dim_Geography, Dim_Job, Dim_Source) with surrogate keys.	Must
FR-04	The system shall define one-to-many, single-direction relationships from each dimension to the fact table.	Must
FR-05	The system shall provide DAX measures for total jobs affected, displacement rate, average automation risk, average AI exposure, average salary, salary change and year-over-year change.	Must
FR-06	The system shall provide four report pages: Executive Summary, Industry Analysis, Risk & Automation, Skills & Training.	Must
FR-07	The system shall provide slicers for year, industry, country/region, job category and risk level, with cross-filtering between visuals.	Must

ID	Requirement	Priority
FR-08	The system shall support drill-down hierarchies (Industry → Company; Region → Country; Job Category → Job Title) and a drill-through detail page.	Should
FR-09	The system shall show tooltips and conditional formatting that highlight high-risk values.	Should
FR-10	The system shall allow export of visuals/pages (PDF, PowerPoint) and underlying data (CSV).	Could
FR-11	The system shall support refresh after the CSV is replaced, without rebuilding the model.	Must

4.1.8 Non-Functional Requirements

Category	Requirement	Design response
Performance	Pages should respond to filter changes within about 2–3 seconds on a standard laptop; full refresh should complete in a few minutes.	Import mode with VertiPaq compression; star schema; measures instead of calculated columns; limited visuals per page.
Security	The report must not expose personal or sensitive data and must be shareable with controlled access.	Synthetic data with no personal identifiers; file-level access control; Row-Level Security can be added if published.
Usability	A first-time user should reach any page and apply a filter without training.	Consistent navigation bar, visible slicers, clear titles, tooltips, reset-filters button.
Accessibility	Content must be readable by users with common visual impairments.	High-contrast palette, minimum font sizes, alt text on visuals, no color-only encoding.
Data integrity	Every fact row must map to exactly one member of each dimension.	Surrogate keys, referential-integrity checks, duplicate removal in Power Query.

Category	Requirement	Design response
Maintainability	Another analyst must be able to rebuild or extend the model.	Modular layers, named Power Query steps, documented measures, display folders.
Scalability	The model should tolerate larger datasets than 30,000 rows.	Star schema and Import mode scale to millions of rows; incremental refresh available in Power BI Service.
Portability	The report must open on any Windows machine with Power BI Desktop.	Single .pbix file; CSV source in a shared, relative or parameterized path.

4.1.9 Software Architecture

The system follows a **Modular Business Intelligence Architecture**: a layered pipeline in which each layer has one responsibility, a defined input and output, and can be replaced without changing the others.

```
flowchart TB
  subgraph L1["Layer 1 – Data Generation"]
    PY["Python script + Claude<br/>synthetic dataset design"]
    CSV[("AI_Jobs_Market.csv<br/>30,000 rows x 25 columns")]
    PY --> CSV
  end
  subgraph L2["Layer 2 – ETL / Preparation"]
    XL["Excel<br/>inspection and profiling"]
    PQ["Power Query<br/>clean, split, key"]
    XL --> PQ
  end
  subgraph L3["Layer 3 – Semantic Model"]
    STAR["Star Schema<br/>1 fact + 4 dimensions"]
    DAX["DAX measures<br/>and hierarchies"]
    STAR --> DAX
  end
  subgraph L4["Layer 4 – Presentation"]
    P1["Executive Summary"]
    P2["Industry Analysis"]
    P3["Risk and Automation"]
    P4["Skills and Training"]
```

```
  end
  subgraph L5["Layer 5 – Distribution"]
    PBIX[".pbix file sharing"]
    SVC["Power BI Service<br/>(optional)"]
  end
  CSV --> XL
  PQ --> STAR
  DAX --> P1
  DAX --> P2
  DAX --> P3
  DAX --> P4
  P1 --> PBIX
  P2 --> PBIX
  P3 --> PBIX
  P4 --> PBIX
  PBIX --> SVC
```

Layer	Components	Responsibility	Output passed on
1. Data Generation	Python, Claude	Design columns, distributions and business rules; generate the synthetic CSV	Raw CSV file
2. ETL / Preparation	Excel, Power Query	Profile data; clean; split into fact and dimensions; create surrogate keys	Clean, keyed tables
3. Semantic Model	Power BI model view, DAX	Star Schema relationships, hierarchies, measures, formatting	Queryable model
4. Presentation	Power BI report canvas	Four interactive pages with slicers, KPIs and charts	Report (.pbix)
5. Distribution	File share, Power BI Service	Share with team and stakeholders; scheduled refresh when published	Accessible report

Why this style. A layered BI architecture separates data preparation from analytics and presentation, which keeps each concern testable, allows the CSV source to be replaced by a database or API later, and matches the way Power BI itself is organized (Power Query

→ model → report). Interactions are one-directional: each layer consumes only the output of the layer beneath it.

4.2 Database Design & Data Modeling

The analytical model is a **Star Schema** with one central fact table, Fact_AI_Job_Market, and four dimension tables: Dim_Company, Dim_Geography, Dim_Job and Dim_Source. The 25-column flat CSV is split so that descriptive attributes live in dimensions and numeric measures live in the fact table.

4.2.1 Entity-Relationship Diagram

Each dimension has a one-to-many relationship to the fact table: one company, geography, job or source can appear in many fact records, and each fact record references exactly one member of each dimension.

```
erDiagram
  DIM_COMPANY ||--o{ FACT_AI_JOB_MARKET : "reports"
  DIM_GEOGRAPHY ||--o{ FACT_AI_JOB_MARKET : "locates"
  DIM_JOB ||--o{ FACT_AI_JOB_MARKET : "describes"
  DIM_SOURCE ||--o{ FACT_AI_JOB_MARKET : "provides"
  DIM_COMPANY {
    int Company_ID PK
    string Company_Name
    string Industry
    string Company_Size_Category
    string AI_Maturity_Level
  }
  DIM_GEOGRAPHY {
    int Geography_ID PK
    string Country
    string Region
    string Continent
  }
  DIM_JOB {
    int Job_ID PK
    string Job_Title
    string Job_Category
    string Skill_Category
    string Required_Skill_Level
    string Education_Level
```

```
}
DIM_SOURCE {
  int Source_ID PK
  string Source_Name
  string Source_Type
  string Reliability_Rating
}
FACT_AI_JOB_MARKET {
  int Record_ID PK
  int Company_ID FK
  int Geography_ID FK
  int Job_ID FK
  int Source_ID FK
  int Report_Year
  string Report_Quarter
  int Estimated_Jobs_Affected_Count
  int Total_Company_Employees
  decimal Avg_Salary_USD
  decimal Salary_Change_Pct
  decimal AI_Exposure_Score
  decimal Automation_Risk_Score
  decimal AI_Adoption_Rate_Pct
  string Displacement_Risk_Level
}
```

4.2.2 Logical Schema

Grain of the fact table: one record per company, job role, geography and data source observation in a given reporting period (30,000 records). All measures are stated at this grain.

Table	Type	Business meaning	Primary key	Foreign keys
Fact_AI_Job_Market	Fact	Measured AI impact on jobs for one observation	Record_ID	Company_ID, Geography_ID, Job_ID, Source_ID
Dim_Company	Dimension	Employer and its industry and size	Company_ID	none

Table	Type	Business meaning	Primary key	Foreign keys
Dim_Geography	Dimension	Location of the observation	Geography_ID	none
Dim_Job	Dimension	Occupation, its category and skill profile	Job_ID	none
Dim_Source	Dimension	Origin and reliability of the observation	Source_ID	none

Relationship	Cardinality	Cross-filter direction	Active
Dim_Company[Company_ID] → Fact[Company_ID]	1 : *	Single (dimension to fact)	Yes
Dim_Geography[Geography_ID] → Fact[Geography_ID]	1 : *	Single	Yes
Dim_Job[Job_ID] → Fact[Job_ID]	1 : *	Single	Yes
Dim_Source[Source_ID] → Fact[Source_ID]	1 : *	Single	Yes

Report_Year and Report_Quarter are kept in the fact table as degenerate time attributes. A dedicated Dim_Date can be added later if finer time intelligence is needed.

4.2.3 Physical Schema

Data types follow Power BI / Power Query types. Keys are integer surrogate keys generated in Power Query; they do not exist in the raw CSV.

Fact_AI_Job_Market

Column	Data type	Key	Description
Record_ID	Whole number	PK	Unique row identifier (1–30,000)

Column	Data type	Key	Description
Company_ID	Whole number	FK	References Dim_Company
Geography_ID	Whole number	FK	References Dim_Geography
Job_ID	Whole number	FK	References Dim_Job
Source_ID	Whole number	FK	References Dim_Source
Report_Year	Whole number		Year of observation (2024–2026)
Report_Quarter	Text		Quarter label (Q1–Q4)
Estimated_Jobs_Affected_Count	Whole number		Estimated jobs displaced or transformed by AI
Total_Company_Employees	Whole number		Employee headcount of the company
Avg_Salary_USD	Fixed decimal		Average salary for the job in USD
Salary_Change_Pct	Decimal		Percentage change in salary attributed to AI adoption
AI_Exposure_Score	Decimal		Degree to which the job's tasks are exposed to AI
Automation_Risk_Score	Decimal		Likelihood the job is automated
AI_Adoption_Rate_Pct	Decimal		Share of the company's processes using AI
Displacement_Risk_Level	Text		Banded category (Low / Medium / High) derived from the risk score

Dim_Company

Column	Data type	Key	Description
Company_ID	Whole number	PK	Surrogate key
Company_Name	Text		Company name (synthetic)
Industry	Text		Industry sector
Company_Size_Category	Text		Small / Medium / Large / Enterprise
AI_Maturity_Level	Text		Level of AI adoption maturity

Dim_Geography

Column	Data type	Key	Description
Geography_ID	Whole number	PK	Surrogate key
Country	Text		Country name
Region	Text		Sub-continental or economic region
Continent	Text		Continent

Dim_Job

Column	Data type	Key	Description
Job_ID	Whole number	PK	Surrogate key
Job_Title	Text		Occupation title
Job_Category	Text		Occupational family (for example technical, administrative, creative)
Skill_Category	Text		Dominant skill group required
Required_Skill_Level	Text		Basic / Intermediate / Advanced
Education_Level	Text		Typical minimum education

Dim_Source

Column	Data type	Key	Description
Source_ID	Whole number	PK	Surrogate key
Source_Name	Text		Name of the data source
Source_Type	Text		Survey, report, job posting, administrative
Reliability_Rating	Text		Reliability grade of the source

Design note: the attribute names above are a proposed allocation of the 25 source columns (10 fact-level, 15 dimension-level). Rename them to match the final CSV headers if any differ.

4.2.4 Normalization Considerations

- **Fact table:** in Third Normal Form. Every non-key column depends on the whole key (Record_ID) and on nothing else; descriptive text was moved out to dimensions.
- **Dimension tables:** deliberately kept denormalized (equivalent to 2NF). For example, Industry stays inside Dim_Company rather than in a separate Industry table. This is the defining trade-off of a star schema: a little redundancy in small tables in exchange for fewer joins and faster, simpler DAX.
- **Why not a snowflake:** splitting Industry, Country or Job_Category into further tables would add relationships and filter paths without a measurable benefit at 30,000 rows.
- **Referential integrity:** every foreign key in the fact table must exist in its dimension; Power Query de-duplicates dimension rows and assigns keys before the fact table is built.
- **Filter direction:** single-direction relationships avoid ambiguous filter paths and keep DAX evaluation predictable.
- **Measures vs columns:** ratios and averages are DAX measures, not stored columns, so they recalculate correctly under any filter context.

4.2.5 Calculation Layer (DAX Measures)

Measure	Purpose
Total Jobs Affected	Sum of estimated jobs affected
Total Employees	Sum of company employees at record grain

Measure	Purpose
Displacement Rate %	Jobs affected as a share of employees
Avg Automation Risk	Mean automation risk score
Avg AI Exposure	Mean AI exposure score
High Risk Share %	Share of records in the High risk band
Avg Salary USD	Mean salary
YoY Jobs Affected %	Year-over-year change in jobs affected

```
Total Jobs Affected =
SUM ( Fact_AI_Job_Market[Estimated_Jobs_Affected_Count] )

Total Employees =
SUM ( Fact_AI_Job_Market[Total_Company_Employees] )

Displacement Rate % =
DIVIDE ( [Total Jobs Affected], [Total Employees] )

Avg Automation Risk =
AVERAGE ( Fact_AI_Job_Market[Automation_Risk_Score] )

Avg AI Exposure =
AVERAGE ( Fact_AI_Job_Market[AI_Exposure_Score] )

High Risk Share % =
DIVIDE (
    CALCULATE (
        COUNTROWS ( Fact_AI_Job_Market ),
        Fact_AI_Job_Market[Displacement_Risk_Level] = "High"
    ),
    COUNTROWS ( Fact_AI_Job_Market )
)

Avg Salary USD =
AVERAGE ( Fact_AI_Job_Market[Avg_Salary_USD] )

YoY Jobs Affected % =
VAR CurrentYear = MAX ( Fact_AI_Job_Market[Report_Year] )
VAR PreviousValue =
```

```
    CALCULATE (
        [Total Jobs Affected],
        FILTER (
            ALL ( Fact_AI_Job_Market[Report_Year] ),
            Fact_AI_Job_Market[Report_Year] = CurrentYear - 1
        )
    )
RETURN
    DIVIDE ( [Total Jobs Affected] - PreviousValue, PreviousValue )
```

Because Total_Company_Employees repeats on every record of a company, it should be summed only at the record grain used for Displacement Rate %. To report a company's headcount on its own, use a distinct-company average or maximum instead.

4.3 Data Flow & System Behavior

This section shows how raw CSV data moves through the system and how its components behave at run time. Notation: in the data flow diagrams, rectangles are external entities, circles are processes, and cylinders are data stores.

4.3.1 Data Flow Diagrams (DFD)

Level 0 – Context diagram. The whole system is one process that exchanges data with three external entities.

```
flowchart LR
  GEN["Data Generator<br/>Python + Claude"]
  ANA["Project Analyst"]
  USR["Dashboard Users<br/>Job Seeker, HR, Policy Maker"]
  SYS(("0<br/>AI Impact on Jobs Market<br/>BI System"))
  GEN -- "raw CSV dataset" --> SYS
  ANA -- "transformation rules, model and DAX definitions, refresh command" -
-> SYS
  SYS -- "validation results, model status" --> ANA
  USR -- "filter and drill selections" --> SYS
  SYS -- "KPIs, charts, tables, exports" --> USR
```

Level 1 – Main processes. The system decomposes into five processes and four data stores.

```
flowchart LR
  GEN["Data Generator"]
  ANA["Project Analyst"]
  USR["Dashboard Users"]
  P1(("1.0<br/>Ingest Raw Data"))
  P2(("2.0<br/>Clean and Transform"))
  P3(("3.0<br/>Build Star Schema"))
  P4(("4.0<br/>Define DAX Measures"))
  P5(("5.0<br/>Render Dashboard"))
  D1[("D1 Raw CSV")]
  D2[("D2 Staging Tables")]
  D3[("D3 Star Schema Model")]
  D4[("D4 Measure Library")]
  GEN -- "CSV file" --> D1
  D1 -- "raw records" --> P1
  ANA -- "refresh command" --> P1
  P1 -- "typed raw table" --> P2
  ANA -- "cleaning rules" --> P2
  P2 -- "clean records" --> D2
  D2 -- "clean records" --> P3
  P3 -- "fact and dimension tables, keys" --> D3
  ANA -- "measure definitions" --> P4
  D3 -- "table structure" --> P4
  P4 -- "measures" --> D4
  D3 -- "filtered data" --> P5
  D4 -- "calculated values" --> P5
  USR -- "filters and drill actions" --> P5
  P5 -- "visuals and KPIs" --> USR
  P3 -- "integrity check results" --> ANA
```

Level 2 – Process 2.0 and 3.0 in detail (ETL in Power Query).

```
flowchart LR
  P1["1.0 Ingest Raw Data"]
  S1(("2.1<br/>Set data types"))
  S2(("2.2<br/>Trim and standardize text"))
  S3(("2.3<br/>Handle nulls and outliers"))
  S4(("2.4<br/>Remove duplicates"))
  S5(("2.5<br/>Derive risk band"))
  D2[("D2 Staging Tables")]
  S6(("3.1<br/>Extract distinct dimension rows"))
  S7(("3.2<br/>Assign surrogate keys"))
  S8(("3.3<br/>Build fact with foreign keys"))
  D3[("D3 Star Schema Model")]
  P1 --> S1 --> S2 --> S3 --> S4 --> S5 --> D2
  D2 --> S6 --> S7 --> S8 --> D3
```

Data flow	From	To	Content
Raw CSV	Data Generator	D1	30,000 rows, 25 columns
Typed raw table	1.0	2.0	Columns with detected types
Clean records	2.0	D2	Standardized, de-duplicated records with risk band
Fact and dimension tables	3.0	D3	Five keyed tables
Measures	4.0	D4	DAX expressions
Filtered data / calculated values	D3, D4	5.0	Aggregates in the current filter context
Filters and drill actions	Dashboard Users	5.0	Slicer and click selections

4.3.2 Sequence Diagrams

Sequence 1 – Filtering and DAX evaluation. A slicer selection changes the filter context; the model propagates it from the dimension to the fact table and the DAX engine recalculates each visual's measures.

```
sequenceDiagram
  actor U as User
  participant S as Slicer
  participant R as Report Canvas
  participant M as Semantic Model
  participant D as DAX Engine
  participant V as VertiPaq Storage
  U->>S: Select Industry = Finance, Year = 2025
  S->>R: Update filter context
  R->>M: Request data for each visual
  M->>D: Evaluate measures in new filter context
  D->>V: Filter Dim_Company and Fact by Report_Year
  Note over D,V: Filter flows from dimension to fact through the 1-to-many
relationship
  V-->>D: Matching fact rows and aggregates
  D-->>M: Measure results
  M-->>R: Result sets
  R-->>U: Re-render KPI cards and charts
```

Sequence 2 – Data refresh. The analyst replaces the CSV and refreshes; Power Query reruns every step and the model reloads.

```
sequenceDiagram
  actor A as Project Analyst
  participant PB as Power BI Desktop
  participant PQ as Power Query Engine
  participant F as CSV File
  participant M as Semantic Model
  A->>PB: Click Refresh
  PB->>PQ: Run query steps
  PQ->>F: Read file
  F-->>PQ: 30,000 raw rows
  PQ->>PQ: Type, clean, de-duplicate, build dimensions and keys
  PQ-->>PB: Five tables
  PB->>M: Load tables, rebuild relationships
  M-->>PB: Model ready
  PB-->>A: Report shows updated data
```

4.3.3 Activity Diagrams

Activity 1 – Analyst workflow (build and refresh the model).

```
flowchart TD
  A(("Start")) --> B["Generate synthetic CSV with Python and Claude"]
  B --> C["Load CSV into Power Query"]
  C --> D["Profile columns: types, nulls, distributions"]
  D --> E{"Data quality acceptable?"}
  E -- "No" --> F["Add cleaning steps"]
  F --> D
  E -- "Yes" --> G["Create dimension tables and surrogate keys"]
  G --> H["Build fact table with foreign keys"]
  H --> I["Create relationships in model view"]
  I --> J["Write DAX measures and hierarchies"]
  J --> K["Design four report pages"]
  K --> L["Run tests and validation"]
  L --> M{"All tests passed?"}
  M -- "No" --> N["Fix model, measures or visuals"]
  N --> L
  M -- "Yes" --> O["Save .pbix and share with team"]
  O --> P(("End"))
```

Activity 2 – Dashboard user workflow.

```
flowchart TD
  A(("Start")) --> B["Open report"]
  B --> C["Choose page from navigation bar"]
  C --> D["Apply slicers: year, industry, country, job category"]
  D --> E["Visuals recalculate"]
  E --> F{"Need more detail?"}
  F -- "Yes" --> G["Drill down or drill through"]
  G --> E
  F -- "No" --> H{"Compare another view?"}
  H -- "Yes" --> C
  H -- "No" --> I{"Export insight?"}
  I -- "Yes" --> J["Export visual or page"]
  I -- "No" --> K(("End"))
  J --> K
```

4.3.4 State Diagrams

State 1 – Lifecycle of a data record. A record moves from raw CSV text to a validated, published model row; failed validation sends it back to cleaning, and a refresh restarts the cycle.

```
stateDiagram-v2
  [*] --> Raw
  Raw --> Loaded : Get Data
  Loaded --> Profiled : Column profiling
  Profiled --> Cleaned : Apply transformation steps
  Cleaned --> Split : Create fact and dimension tables
  Split --> Modeled : Keys and relationships defined
  Modeled --> Validated : Tests pass
  Modeled --> Cleaned : Integrity or value defect found
  Validated --> Published : Report saved and shared
  Published --> Loaded : Refresh with new CSV
  Published --> [*]
```

State 2 – Filter state of a report page. The page moves between unfiltered, filtered, cross-filtered and drilled states and returns to the default state on reset.

```
stateDiagram-v2
  [*] --> Unfiltered
  Unfiltered --> Filtered : Slicer selected
  Filtered --> CrossFiltered : Visual element clicked
  CrossFiltered --> Filtered : Click again to deselect
  Filtered --> Drilled : Drill down or drill through
  CrossFiltered --> Drilled : Drill down or drill through
  Drilled --> Filtered : Drill up or back
  Filtered --> Unfiltered : Clear slicers
  CrossFiltered --> Unfiltered : Reset filters button
  Drilled --> Unfiltered : Reset filters button
```

4.3.5 Class Diagram

The class diagram models the logical classes of the system: the data pipeline, the star schema entities, the semantic model and the report layer.

```
classDiagram
  class CsvSource {
    +String filePath
    +int rowCount
    +int columnCount
    +load() DataTable
  }
  class PowerQueryETL {
    +setTypes() void
```

```
    +cleanText() void
    +handleNulls() void
    +removeDuplicates() void
    +deriveRiskBand() void
    +buildDimensions() void
    +assignKeys() void
    +buildFact() void
  }
  class DimensionTable {
    <<abstract>>
    +int surrogateKey
    +getAttributes() List
  }
  class DimCompany {
    +String Company_Name
    +String Industry
    +String Company_Size_Category
    +String AI_Maturity_Level
  }
  class DimGeography {
    +String Country
    +String Region
    +String Continent
  }
  class DimJob {
    +String Job_Title
    +String Job_Category
    +String Skill_Category
    +String Required_Skill_Level
    +String Education_Level
  }
  class DimSource {
    +String Source_Name
    +String Source_Type
    +String Reliability_Rating
  }
  class FactAIJobMarket {
    +int Record_ID
    +int Report_Year
    +String Report_Quarter
    +int Estimated_Jobs_Affected_Count
    +int Total_Company_Employees
    +decimal Avg_Salary_USD
```

```
    +decimal Salary_Change_Pct
    +decimal AI_Exposure_Score
    +decimal Automation_Risk_Score
    +decimal AI_Adoption_Rate_Pct
    +String Displacement_Risk_Level
  }
  class SemanticModel {
    +List relationships
    +refresh() void
    +evaluate(measure, filterContext) Value
  }
  class DaxMeasure {
    +String name
    +String expression
    +calculate(filterContext) Value
  }
  class ReportPage {
    +String title
    +render() void
    +resetFilters() void
  }
  class Slicer {
    +String field
    +List selectedValues
    +applyFilter() void
  }
  class Visual {
    +String type
    +DaxMeasure measure
    +refreshData() void
  }
  CsvSource --> PowerQueryETL : feeds
  PowerQueryETL ..> FactAIJobMarket : creates
  PowerQueryETL ..> DimensionTable : creates
  DimensionTable <|-- DimCompany
  DimensionTable <|-- DimGeography
  DimensionTable <|-- DimJob
  DimensionTable <|-- DimSource
  DimensionTable "1" --> "0..*" FactAIJobMarket : filters
  SemanticModel o-- FactAIJobMarket
  SemanticModel o-- DimensionTable
  SemanticModel o-- DaxMeasure
```

```
  ReportPage *-- Visual
  ReportPage *-- Slicer
  Visual ..> DaxMeasure : displays
  Slicer ..> SemanticModel : sets filter context
```

4.4 UI/UX Design & Prototyping

The report uses a 16:9 canvas (1280 × 720 px) with a consistent structure on every page: a navigation bar at the top, a slicer strip beneath it, a row of KPI cards, then two rows of analytical visuals. Consistency lets users learn the layout once and apply it to all four pages.

4.4.1 Wireframes & Mockups

Common page template

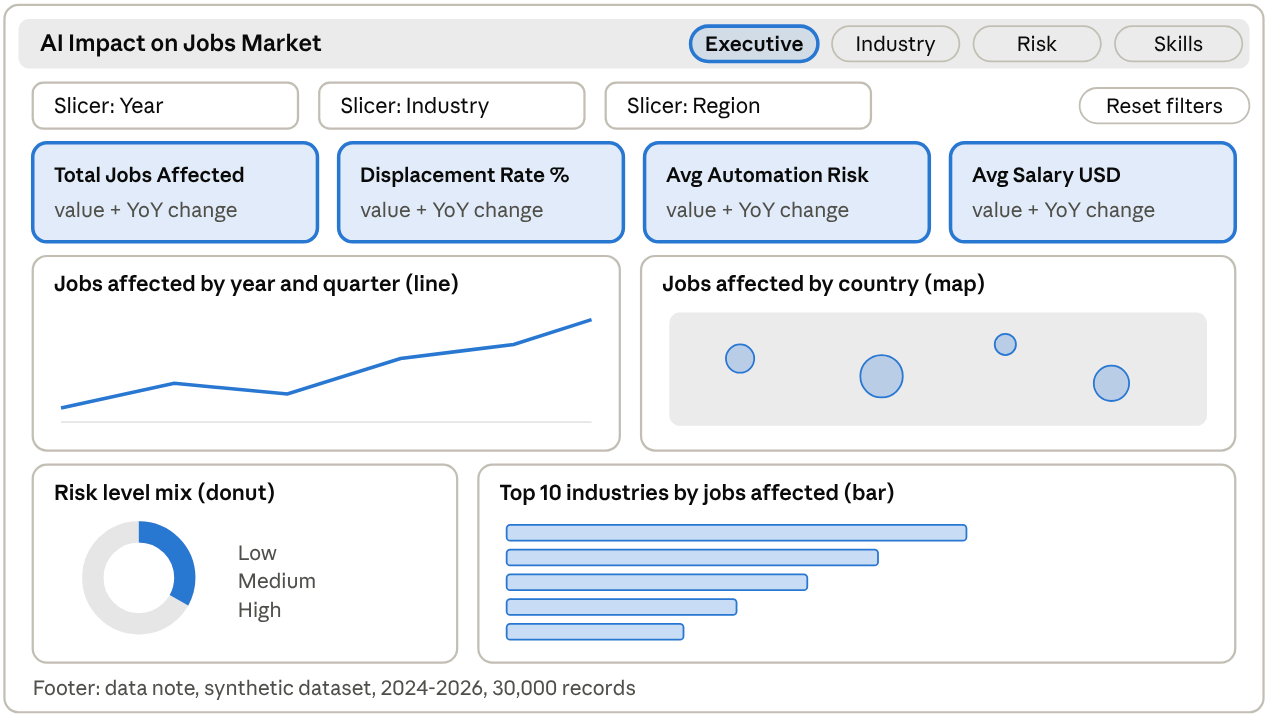
Zone	Position	Content
Header	Top, full width	Report title, page title, navigation buttons for the four pages, reset-filters button
Slicer strip	Below header	Page-specific slicers; year and region slicers are synchronized across pages
KPI row	Below slicers	Three to four KPI cards with value and short label
Analysis area	Middle	Two rows of two or three visuals
Footer	Bottom	Data note: synthetic dataset, 2024–2026, 30,000 records

Page 1 – Executive Summary. Purpose: give any stakeholder the headline picture in under a minute.

Zone	Visual	Fields and measures
Slicers	Year, Industry, Region	Report_Year, Industry, Region
KPI cards	Four cards	Total Jobs Affected, Displacement Rate %, Avg Automation Risk, Avg Salary USD

Zone	Visual	Fields and measures
Trend	Line chart	Total Jobs Affected by Report_Year and Report_Quarter
Geography	Filled map or bar chart	Total Jobs Affected by Country
Risk mix	Donut chart	Record count by Displacement_Risk_Level
Ranking	Horizontal bar chart	Top 10 industries by Total Jobs Affected

Executive Summary: headline KPIs first, then trend, map, risk mix and ranking



Executive Summary wireframe · header, slicers, 4 KPI cards, 4 visuals

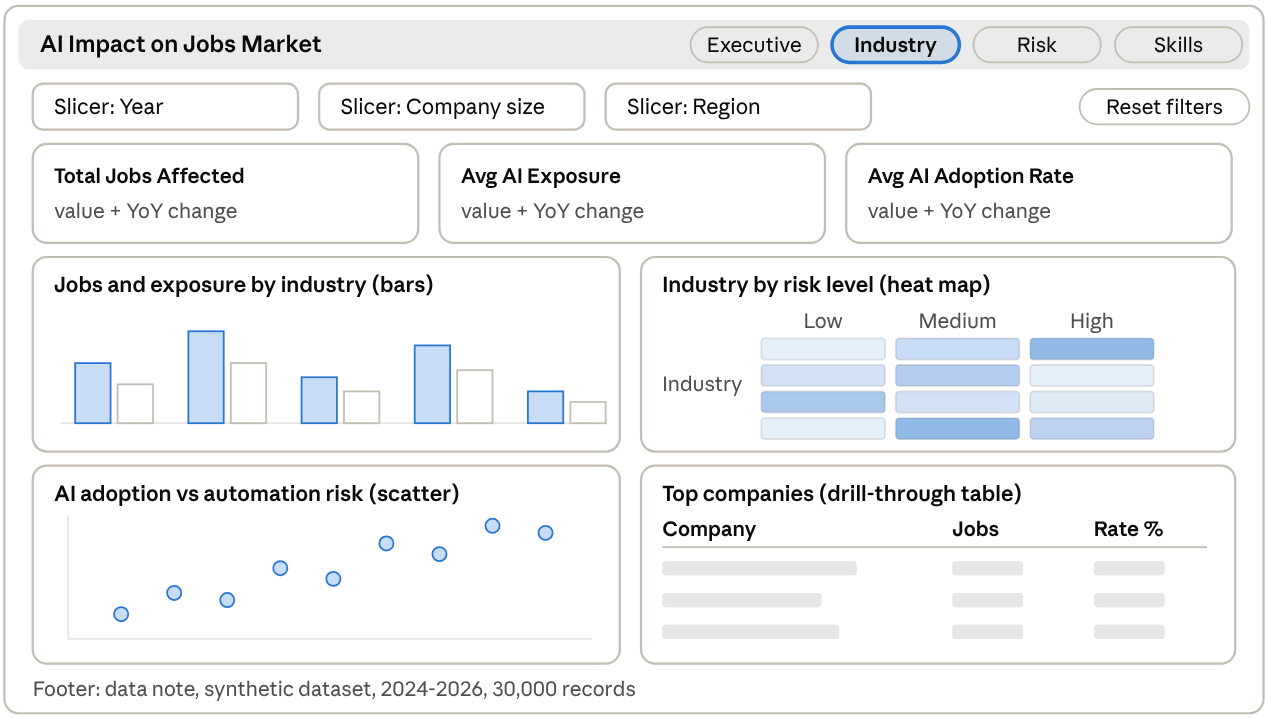
The KPI cards (highlighted) sit directly under the slicers so the headline numbers are read first; trend, geography, risk mix and ranking follow.

Page 2 – Industry Analysis. Purpose: compare industries and company sizes.

Zone	Visual	Fields and measures
Slicers	Year, Company Size, Region	Report_Year, Company_Size_Category, Region

Zone	Visual	Fields and measures
KPI cards	Three cards	Total Jobs Affected, Avg AI Exposure, AI adoption (average of AI_Adoption_Rate_Pct)
Comparison	Clustered bar chart	Total Jobs Affected and Avg AI Exposure by Industry
Heat map	Matrix with conditional formatting	Industry (rows) by Displacement_Risk_Level (columns), value = record count
Relationship	Scatter plot	AI_Adoption_Rate_Pct (x) vs Avg Automation Risk (y), one point per industry
Detail	Table with drill-through	Company_Name, Total Jobs Affected, Displacement Rate %

Industry Analysis: compare industries on exposure, risk mix and adoption



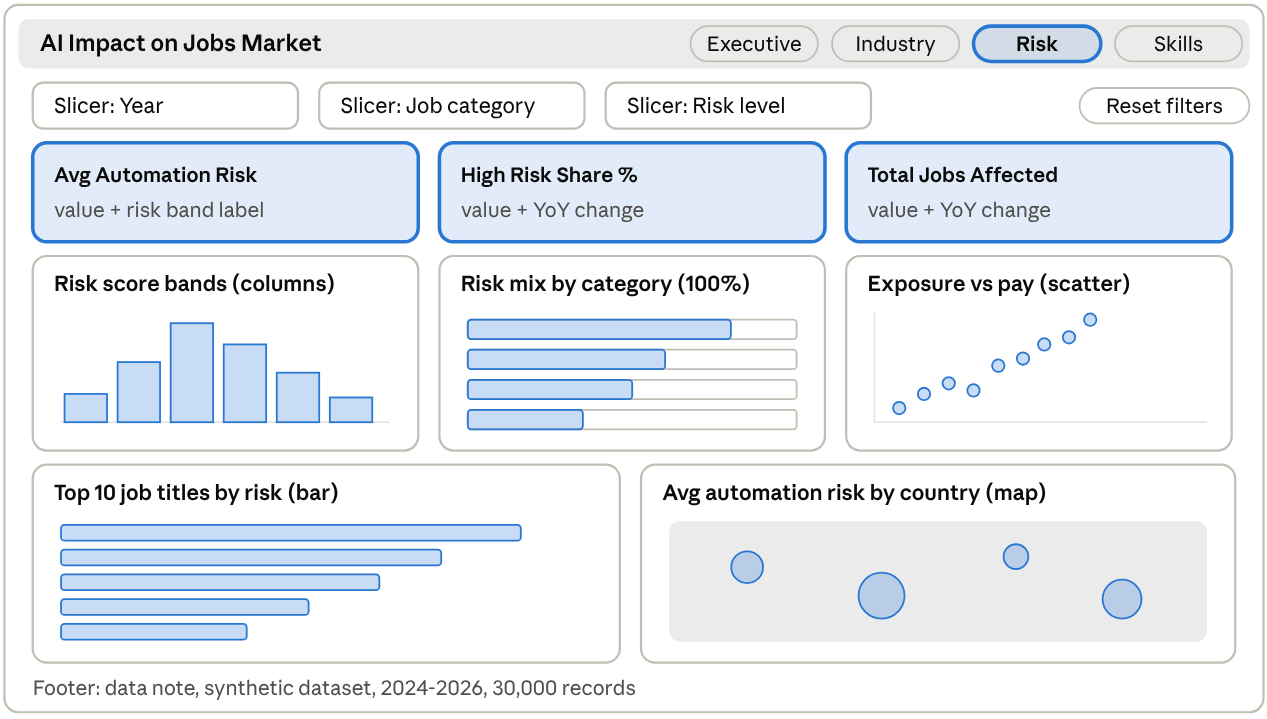
Industry Analysis wireframe · 3 KPI cards, bars, heat map, scatter, table

Industry comparison sits on the top row; the scatter and the drill-through table below let users move from industry to company.

Page 3 – Risk & Automation. Purpose: show which jobs and places carry the highest displacement risk.

Zone	Visual	Fields and measures
Slicers	Year, Job Category, Risk Level	Report_Year, Job_Category, Displacement_Risk_Level
KPI cards	Three cards	Avg Automation Risk, High Risk Share %, Total Jobs Affected
Distribution	Column chart	Record count by Automation_Risk_Score band
Composition	100% stacked bar	Risk level share by Job_Category
Economic impact	Scatter plot	AI_Exposure_Score (x) vs Salary_Change_Pct (y)
Ranking	Bar chart with data labels	Top 10 Job_Title by Avg Automation Risk
Geography	Map	Avg Automation Risk by Country

Risk and Automation: distribution and mix first, then top jobs and countries



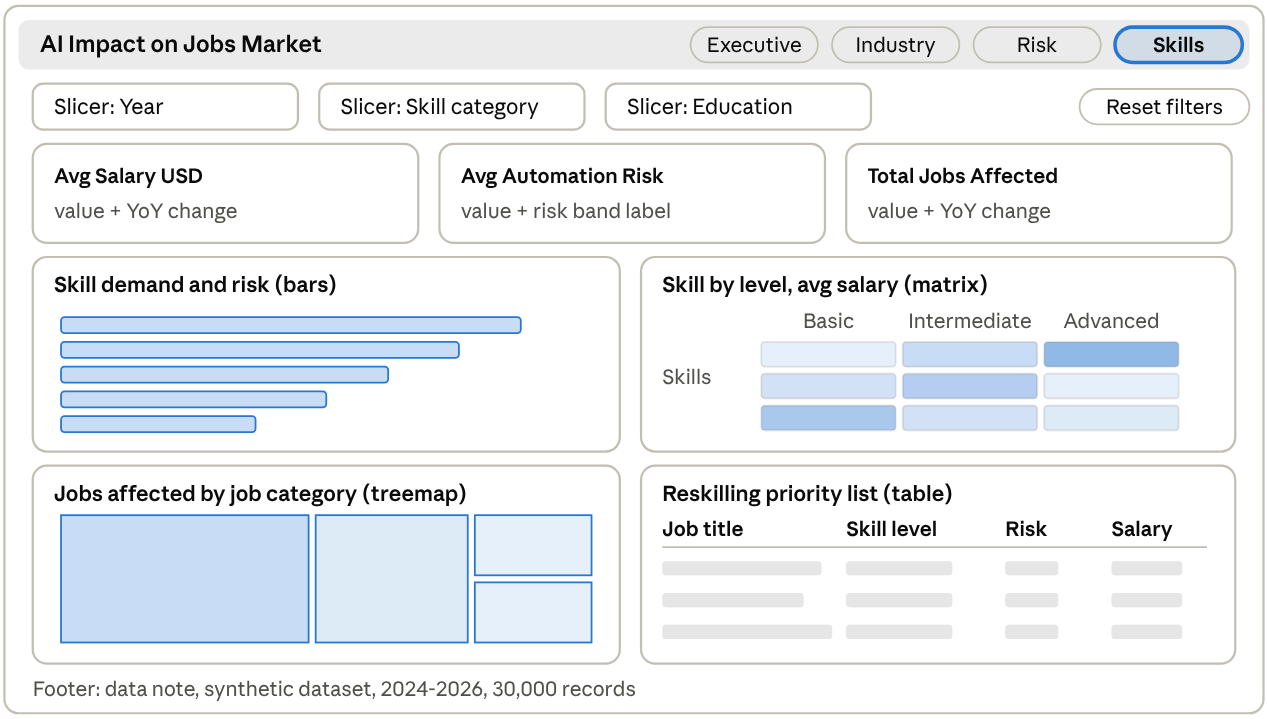
Risk and Automation wireframe · 3 KPI cards, 3 distribution visuals, ranking, map

The top row shows how risk is distributed; the bottom row names the jobs and countries that carry it.

Page 4 – Skills & Training. Purpose: guide reskilling decisions.

Zone	Visual	Fields and measures
Slicers	Year, Skill Category, Education Level	Report_Year, Skill_Category, Education_Level
KPI cards	Three cards	Avg Salary USD, Avg Automation Risk, Total Jobs Affected
Skill demand	Bar chart	Record count and Avg Automation Risk by Skill_Category
Skill matrix	Matrix	Skill_Category (rows) by Required_Skill_Level (columns), value = Avg Salary USD
Job mix	Treemap	Total Jobs Affected by Job_Category
Priority list	Table	Job_Title, Required_Skill_Level, Avg Automation Risk, Avg Salary USD, sorted by risk

Skills and Training: skill demand and levels on top, priority list below



Skills and Training wireframe · 3 KPI cards, bars, matrix, treemap, priority table

Skill demand and level-by-salary views lead; the priority table turns them into a ranked reskilling list.

4.4.2 UI/UX Guidelines

Design principles

- **Clarity first:** one message per visual; every title states what is shown, not just the field name.
- **Progressive disclosure:** summary on page 1, detail on pages 2–4, record-level detail through drill-through.
- **Consistency:** same positions for navigation, slicers and KPI cards on all pages; same color meaning everywhere.
- **Interactivity with control:** cross-filtering on by default, one-click reset, tooltips for context.
- **Low cognitive load:** at most six to eight visuals per page, generous white space, no 3D or decorative effects.

Color scheme (proposed; adjust to final theme file)

Role	Color	Hex	Use
Primary	Navy	#1F3A5F	Header, titles, primary series
Secondary	Teal	#2A9D8F	Secondary series, positive change
Accent	Sky blue	#4C9BE8	Highlights, selected state
Risk Low	Green	#2E8B57	Low risk band
Risk Medium	Amber	#E9A100	Medium risk band
Risk High	Red	#C0392B	High risk band, warnings
Background	Light gray	#F3F5F8	Canvas background
Card	White	#FFFFFF	Visual and KPI card background
Text	Charcoal	#2B2F36	Body text and labels

Typography

Element	Font	Size	Weight
Page title	Segoe UI	22 pt	Bold
Visual title	Segoe UI	14 pt	Semibold
KPI value	Segoe UI	28 pt	Bold
KPI label	Segoe UI	11 pt	Regular
Axis labels, legends, slicers	Segoe UI	10–12 pt	Regular

Card layout. Each visual sits on a white card with 8 px rounded corners, a light border, 16 px padding and 16 px gutters, aligned to an 8 px grid. KPI cards use a large value, a small label beneath and an optional change indicator (arrow plus percentage) for year-over-year movement.

Accessibility considerations

- Text and key graphics meet a contrast ratio of at least 4.5:1 against their background.
- Risk is never encoded by color alone: bands also carry text labels (Low, Medium, High) and data labels.

- Palette is checked for common color-vision deficiencies; red and green are separated by lightness and labeled.
- Minimum font size is 10 pt; alt text is set on every visual; tab order follows reading order so the report can be navigated by keyboard.
- Titles, tooltips and number formats use plain language and consistent units (USD, %, counts with thousands separators).

Interaction design

- Navigation buttons with a highlighted state for the current page.
- Synchronized year and region slicers; page-specific slicers elsewhere.
- A bookmark-based Reset Filters button on every page.
- Custom tooltip page showing risk level, average salary and jobs affected for the hovered item.
- Drill-through from industry, company and job-title visuals to a detail page.

4.5 System Deployment & Integration

4.5.1 Technology Stack

Tier	Technology	Role in the system
Data generation (back end)	Python (for example pandas and NumPy), Claude	Design the schema and business rules, generate the 30,000-record synthetic dataset, export CSV
Data storage	CSV file (UTF-8, comma-delimited)	Single raw source file; later replaceable by a database
Data profiling	Microsoft Excel	Inspect, sample and cross-check raw data and test results
ETL	Power Query (M language)	Type conversion, cleaning, de-duplication, dimension extraction, key assignment
Data modeling (database / BI engine)	Power BI Desktop model view, VertiPaq in-memory engine	Star Schema, relationships, hierarchies, compressed in-memory storage

Tier	Technology	Role in the system
Calculation	DAX	Measures, ratios, year-over-year comparison
Front end	Power BI report canvas	Four interactive pages, slicers, tooltips, drill-through, theme
Distribution	.pbix file on shared storage; Power BI Service (optional)	Team sharing, publishing, scheduled refresh

4.5.2 Deployment Diagram

The system runs on the analyst's workstation and is shared as a single report file. Publishing to Power BI Service is optional and shown with dashed lines.

```
flowchart TB
  subgraph WS["Node: Analyst Workstation (Windows PC)"]
    subgraph PYENV["Execution environment: Python runtime"]
      GENPY["generate_dataset.py"]
    end
    CSVF[("AI_Jobs_Market.csv")]
    subgraph EXCELENV["Execution environment: Microsoft Excel"]
      PROF["Profiling and test workbook"]
    end
    subgraph PBID["Execution environment: Power BI Desktop"]
      PQE["Power Query engine"]
      VPQ["VertiPaq model and DAX engine"]
      RPT["Report canvas"]
    end
    PBIXF[("AI_Impact_Jobs.pbix")]
  end
  subgraph CLOUDAI["Node: Claude (cloud AI assistant)"]
    CLD["Design and code assistance"]
  end
  subgraph SHARE["Node: Shared storage (OneDrive, SharePoint or network
folder)"]
    SHF[("Versioned .pbix and CSV copies")]
  end
  subgraph CLIENTS["Node: Team and stakeholder devices"]
    PBIC["Power BI Desktop"]
    BRW["Web browser"]
  end
```

```
  subgraph SVC["Node: Power BI Service (optional)"]
    WSP["Workspace: dataset and report"]
  end
  CLD -. "prompts and generated code" .-> GENPY
  GENPY --> CSVF
  CSVF --> PROF
  CSVF --> PQE
  PQE --> VPQ
  VPQ --> RPT
  RPT --> PBIXF
  PBIXF --> SHF
  SHF --> PBIC
  PBIXF -. "publish" .-> WSP
  WSP -. "view" .-> BRW
```

Node	Hardware / environment	Artifacts deployed
Analyst workstation	Windows PC with sufficient RAM for in-memory model	Python script, CSV, Excel workbook, .pbix
Claude	Cloud-hosted AI assistant	No deployed artifact; used during design and generation
Shared storage	Cloud drive or network share	Versioned .pbix and CSV
Team devices	Windows PC with Power BI Desktop, or any device with a browser	Copy of .pbix, or browser session
Power BI Service	Microsoft cloud (optional)	Published dataset and report

4.5.3 Component Diagram

Components are self-contained units with defined interfaces. An arrow reads "depends on".

```
flowchart LR
  C1["Data Generator<br/>Python script"]
  C2["Raw Data File<br/>CSV"]
  C3["Profiling Workbook<br/>Excel"]
  C4["ETL Component<br/>Power Query queries"]
  C5["Data Model<br/>Star Schema"]
  C6["Measure Library<br/>DAX"]
  C7["Report Pages<br/>4 pages"]
  C8["Theme and Navigation<br/>JSON theme, bookmarks"]
  C9["Distribution<br/>.pbix and Power BI Service"]
  C2 --> C1
  C3 --> C2
  C4 --> C2
  C5 --> C4
  C6 --> C5
  C7 --> C6
  C7 --> C8
  C9 --> C7
```

Component	Provided interface	Required interface
Data Generator	Writes CSV with 25 named columns	Business rules and column specification
Raw Data File	Comma-delimited rows with header	None
ETL Component	Five clean, keyed tables	CSV file path (parameter)
Data Model	Relationships, hierarchies, tables	Keyed tables from ETL
Measure Library	Named DAX measures	Data Model
Report Pages	Visuals, slicers, drill-through	Measures, theme
Distribution	Shareable report	Completed report file

4.6 Additional Deliverables

4.6.1 API Documentation and Data Ingestion

This project does **not** use an external live API. Data arrives as a pre-processed CSV file that Power BI imports through Power Query. The interface that matters is therefore the file contract and the ingestion procedure documented below.

Data source contract

Property	Specification
File	AI_Jobs_Market.csv
Format	Plain text, comma-delimited, UTF-8, first row is the header
Size	30,000 records, 25 columns
Period	2024–2026
Origin	Synthetic data generated with Python and Claude; contains no personal data
Connector	Power Query Text/CSV (Csv.Document)
Load mode	Import (data stored in the in-memory model)
Source path	Held in a Power Query parameter (p_FilePath) so the location can change without editing queries

Ingestion procedure

1. Place the CSV in the agreed folder and set p_FilePath.
2. Power Query reads the file and promotes the first row to headers.
3. Column types are applied (whole number, decimal, currency, text).
4. Text is trimmed and standardized; duplicates and invalid rows are removed.
5. Dimension tables are extracted with distinct values and given surrogate keys.
6. The fact table is built by merging back to each dimension to retrieve its key, keeping only keys and measures.
7. Tables load into the model; relationships are validated.

Illustrative Power Query (M) code

```
// Raw_AI_Jobs
let
    Source  = Csv.Document(File.Contents(p_FilePath),
              [Delimiter = ",", Columns = 25, Encoding = 65001, QuoteStyle =
QuoteStyle.Csv]),
    Headers = Table.PromoteHeaders(Source, [PromoteAllScalars = true]),
    Typed   = Table.TransformColumnTypes(Headers, {
                {"Report_Year", Int64.Type},
                {"Estimated_Jobs_Affected_Count", Int64.Type},
                {"Total_Company_Employees", Int64.Type},
                {"Avg_Salary_USD", Currency.Type},
                {"AI_Exposure_Score", type number},
                {"Automation_Risk_Score", type number}}),
    Trimmed = Table.TransformColumns(Typed, {{"Company_Name", Text.Trim, type
text}}),
    Result  = Table.Distinct(Trimmed)
in
    Result

// Dim_Company
let
    Source = Raw_AI_Jobs,
    Cols   = Table.SelectColumns(Source,
             {"Company_Name", "Industry", "Company_Size_Category",
"AI_Maturity_Level"}),
    Dist   = Table.Distinct(Cols),
    Keyed  = Table.AddIndexColumn(Dist, "Company_ID", 1, 1, Int64.Type)
in
    Keyed
```

The column names in the code follow the proposed schema in 4.2.3 and should be aligned with the final CSV headers.

Future integration option. If the project later consumes live data, the same ETL layer can read a REST API with Web.Contents or a database with a native connector, and Power BI Service can refresh it on a schedule. The star schema and report do not change.

4.6.2 Testing & Validation

Expected values are obtained by an **independent calculation** (Excel pivot table or a Python script on the raw CSV) and compared with the Power BI result for the same filter. No measure is accepted on its own output.

Unit tests for DAX measures

ID	Measure	Test condition	Expected result
UT-01	Total Jobs Affected	No filters	Equals the sum of Estimated_Jobs_Affected_Count in the raw CSV
UT-02	Total Jobs Affected	One industry and one year selected	Equals the Excel pivot total for the same selection
UT-03	Total Jobs Affected	Grouped by industry	Industry values add up to the grand total
UT-04	Displacement Rate %	Any selection	Equals Total Jobs Affected divided by Total Employees; blank, not an error, when the denominator is zero
UT-05	Avg Automation Risk	No filters	Equals the average of Automation_Risk_Score in the raw CSV
UT-06	High Risk Share %	Any selection	Between 0% and 100%; High + Medium + Low shares sum to 100%
UT-07	Avg Salary USD	Single job category	Equals the Excel average for that category
UT-08	YoY Jobs Affected %	Year = 2024 (first year)	Blank, because no prior year exists
UT-09	YoY Jobs Affected %	Year = 2025	Equals (value 2025 − value 2024) divided by value 2024
UT-10	All measures	Selection that matches no rows	Return blank, with no error or divide-by-zero

Integration tests for model relationships and data load

ID	Test	Expected result
IT-01	Row count of Fact_AI_Job_Market after refresh	30,000

ID	Test	Expected result
IT-02	Column count of the raw query	25
IT-03	Primary keys unique in each dimension and in the fact table	No duplicate keys
IT-04	Foreign keys in the fact table with no match in a dimension	Zero; no (Blank) member appears in any slicer
IT-05	Relationship properties in model view	Four one-to-many, single-direction, active relationships
IT-06	Filter propagation: select one Industry	Fact record count equals the Excel COUNTIFS result
IT-07	Null check on key and measure columns	No nulls in keys; nulls in measures handled as defined in Power Query
IT-08	Refresh after replacing the CSV with an updated copy	Model loads without error and row count matches the new file

Non-functional checks

Area	Method	Acceptance
Performance	Power BI Performance Analyzer on each page; DAX Studio for slow measures	Each page renders within the 2–3 second target on the reference workstation
Usability	Task-based observation of first-time users	Users complete the UAT scenarios without help
Accessibility	Manual checklist: contrast, alt text, tab order, color-independence	All items pass

User Acceptance Testing (UAT) plan

- **Participants:** at least one representative for each actor: a job seeker, an HR/employer user, a policy-maker role and a second analyst.
- **Environment:** the final .pbix opened in Power BI Desktop on a clean machine.

- **Procedure:** each participant performs the scenarios below, records pass or fail and comments; defects are logged, fixed and retested; sign-off follows when all critical scenarios pass.

ID	Actor	Scenario	Acceptance criterion
UAT-01	Policy Maker	Find total jobs affected and the average automation risk for 2025	Correct values found within one minute
UAT-02	HR / Employer	Compare two industries on exposure and jobs affected	Comparison read correctly from Industry Analysis
UAT-03	Job Seeker	Find the risk level of a chosen job category and the skills it requires	Risk and skill level identified correctly
UAT-04	HR / Employer	Drill from an industry to a company and open the detail page	Detail page shows the selected company only
UAT-05	Any	Apply three slicers, then reset	All visuals return to the default state
UAT-06	Policy Maker	Export a page to PDF	Export is complete and legible
UAT-07	Project Analyst	Replace the CSV and refresh	Report updates without manual repair

4.6.3 Deployment Strategy

Hosting environment. Development and the graduation demonstration run in **Power BI Desktop** from a single .pbix file. For wider access, the report can be published to a workspace in **Power BI Service**, where stakeholders view it in a browser. Sharing inside Power BI Service depends on the licenses held by the people involved.

File-sharing pipeline for the team

1. The analyst saves the working file with a version suffix (for example AI_Impact_Jobs_v1.3.pbix).
2. The file and the source CSV are copied to the shared folder (OneDrive, SharePoint or a network share) that keeps version history.

3. Teammates open the latest version; changes are made on a copy and merged by the analyst to avoid conflicting edits.
4. A short change log records each version: what changed, who tested it, and the date.
5. The final approved version is the one submitted and, if needed, published.

Scaling considerations

Concern	Current state	Scaling path
Data volume	30,000 rows, Import mode	Star schema and column compression handle millions of rows; use incremental refresh in Power BI Service
Data source	Static CSV	Replace with a database or API; only the first Power Query step changes
Refresh	Manual in Desktop	Scheduled refresh after publishing; gateway if the source is on-premises
Users	Small team	Workspace roles and Row-Level Security by region or industry
Model growth	Four dimensions	Add Dim_Date, Dim_Skill or new fact tables as additional conformed dimensions
Governance	File-based versioning	Deployment pipelines and source control for the model definition